

GUESSING NUMBER GAME ALGORITHM

The number guessing game begins by welcoming the player and displaying the rules, which explain the available difficulty levels, the number ranges, the number of attempts allowed, and the overall goal of guessing the correct number before running out of chances. The player is then prompted to choose a difficulty level Easy, Medium, or Hard and the system continues asking until a valid option is selected. Once a valid choice is made, the game sets up the appropriate conditions using that difficulty: it assigns the number range, determines the maximum number of attempts, initializes the attempt counter to zero, sets the win status to false, and generates a random secret number within the selected range.

The game then enters its main loop, where the player repeatedly makes guesses. Before each attempt, the system checks whether the player has already used up all their attempts; if so, the game ends immediately and reveals the correct number. If attempts are still available, the player is asked to enter a guess. The system validates the input by ensuring it is a number and within the allowed range; if not, the player is prompted to try again. Once a valid guess is entered, the attempt counter is increased.

After each valid guess, the system checks whether the player has correctly guessed the secret number. If the guess is correct, a success message is displayed and the game ends immediately. If the guess is incorrect, feedback is provided based on the selected difficulty level: in Easy mode, the player is told whether the guess is too high or too low; in Medium mode, the player is simply told the guess is incorrect; and in Hard mode, the player is told the guess is wrong and may receive an additional penalty that increases the number of attempts used. The game then continues looping until either the player guesses correctly or all attempts are exhausted.

If the player runs out of attempts without guessing the number, the game ends and reveals the correct answer. Finally, the player is given the option to play again. If they choose yes, the game restarts from the beginning; otherwise, the program terminates.